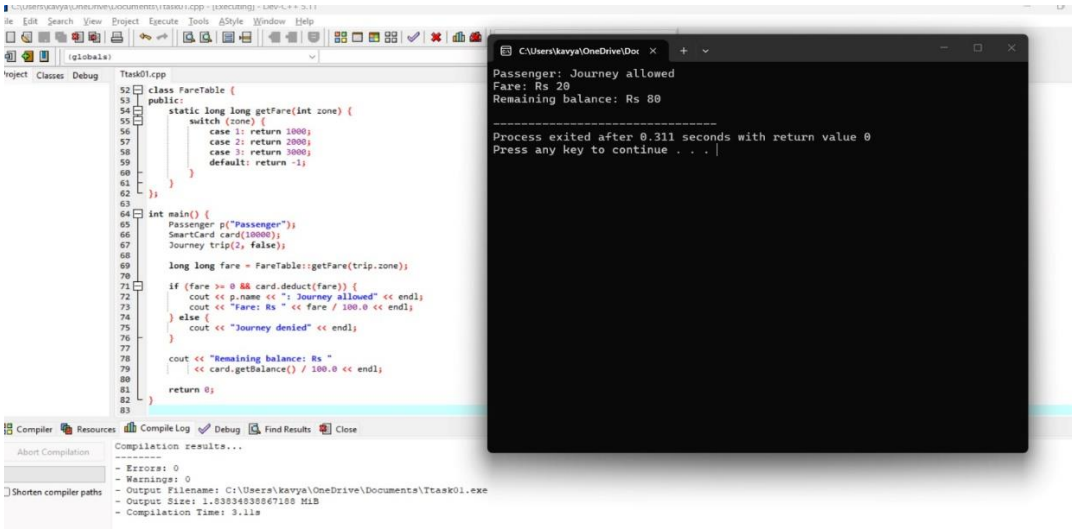
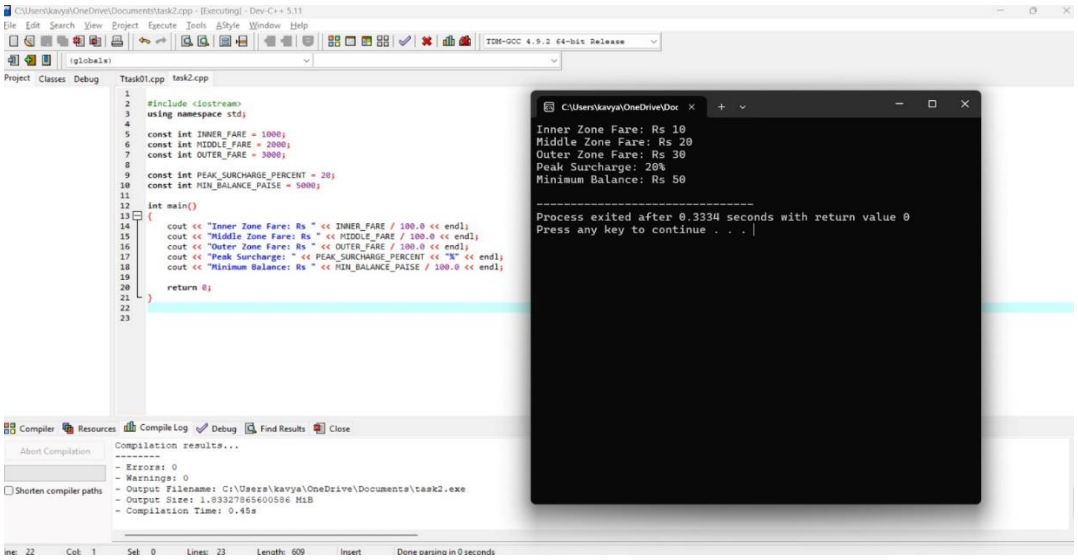
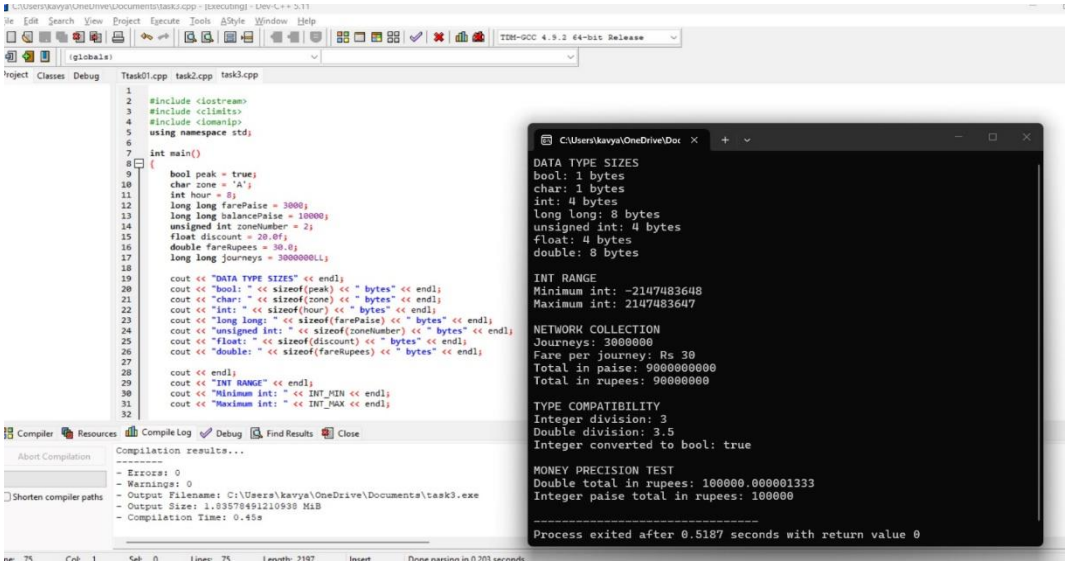
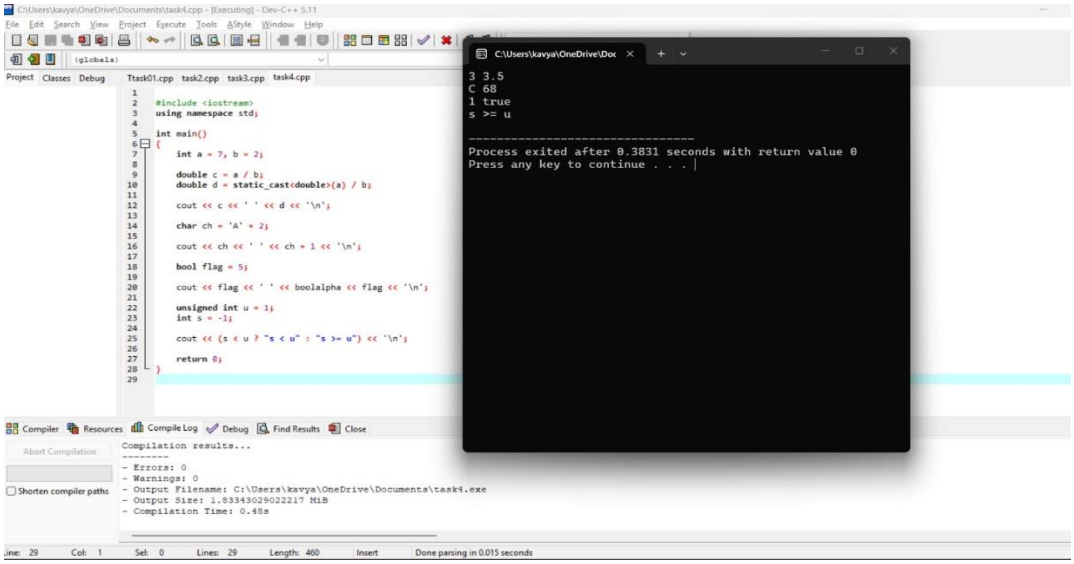
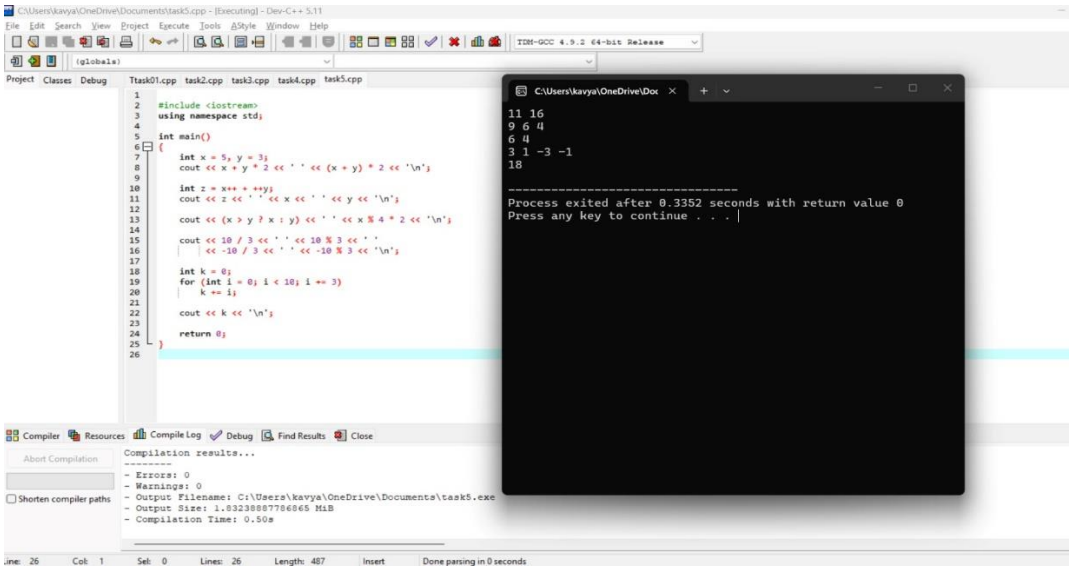
Task 1

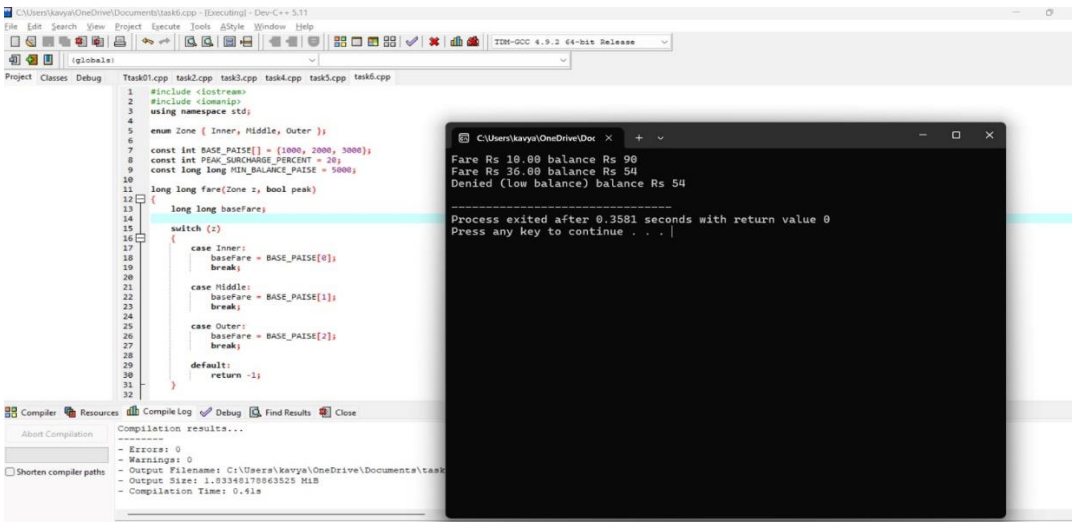

Ttask01.cpp
class FareTable {
public:
static long long getFare(int zone) {
switch (zone) {
case 1: return 1000;
case 2: return 2000;
case 3: return 3000;
default: return -1;
}
}
};
int main() {
Passenger p("Passenger");
SmartCard card(10000);
Journey trip(2, false);
long long fare = FareTable::getFare(trip.zone);
if (fare >= 0 && card.deduct(fare)) {
cout << p.name << ": Journey allowed" << endl;
cout << "Fare: Rs " << fare / 100.0 << endl;
} else {
cout << "Journey denied" << endl;
}
cout << "Remaining balance: Rs "
<< card.getBalance() / 100.0 << endl;
return 0;
}
C:\Users\kavya\OneDrive\Doc
Passenger: Journey allowed
Fare: Rs 20
Remaining balance: Rs 80
Process exited after 0.311 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\Ttask01.exe
- Output Size: 1.83834838867188 MiB
- Compilation Time: 3.11s

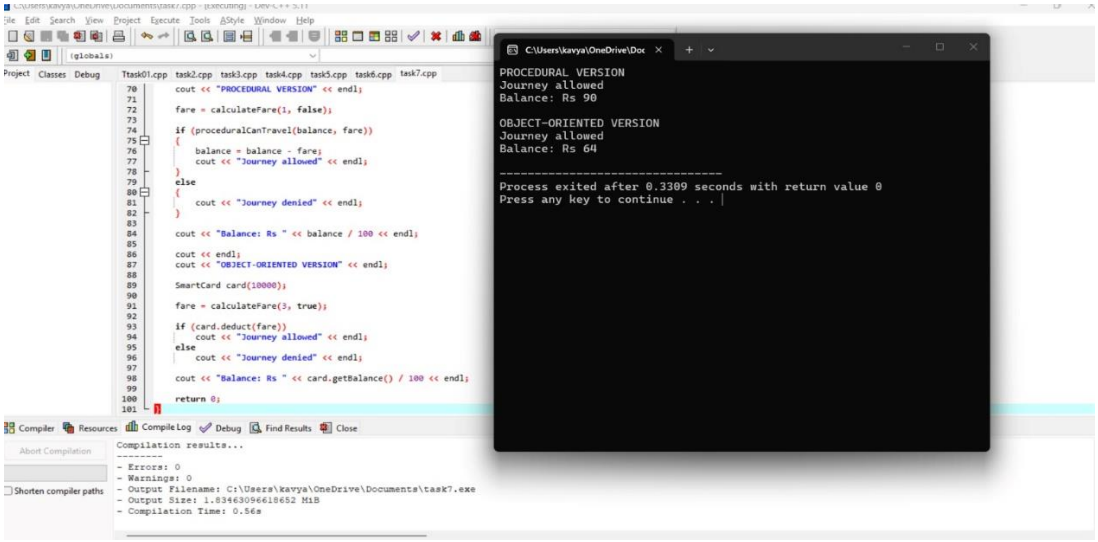

Ttask01.cpp
task2.cpp
#include <iostream>
using namespace std;
const int INNER_FARE = 1000;
const int MIDDLE_FARE = 2000;
const int OUTER_FARE = 3000;
const int PEAK_SURCHARGE_PERCENT = 20;
const int MIN_BALANCE_PAISE = 5000;
int main()
{
cout << "Inner Zone Fare: Rs " << INNER_FARE / 100.0 << endl;
cout << "Middle Zone Fare: Rs " << MIDDLE_FARE / 100.0 << endl;
cout << "Outer Zone Fare: Rs " << OUTER_FARE / 100.0 << endl;
cout << "Peak Surcharge: " << PEAK_SURCHARGE_PERCENT << "%" << endl;
cout << "Minimum Balance: Rs " << MIN_BALANCE_PAISE / 100.0 << endl;
return 0;
}
C:\Users\kavya\OneDrive\Doc
Inner Zone Fare: Rs 10
Middle Zone Fare: Rs 20
Outer Zone Fare: Rs 30
Peak Surcharge: 20%
Minimum Balance: Rs 50
Process exited after 0.3334 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task2.exe
- Output Size: 1.8332786560058 MiB
- Compilation Time: 0.45s


Ttask01.cpp
task2.cpp
task3.cpp
#include <iostream>
#include <climits>
#include <iomanip>
using namespace std;
int main()
{
bool peak = true;
char zone = 'A';
int hour = 8;
long long farePaise = 3000;
long long balancePaise = 10000;
unsigned int zoneNumber = 2;
float discount = 20.0f;
double fareRupees = 30.0;
long long journeys = 3000000LL;
cout << "DATA TYPE SIZES" << endl;
cout << "bool: " << sizeof(peak) << " bytes" << endl;
cout << "char: " << sizeof(zone) << " bytes" << endl;
cout << "int: " << sizeof(hour) << " bytes" << endl;
cout << "long long: " << sizeof(farePaise) << " bytes" << endl;
cout << "unsigned int: " << sizeof(zoneNumber) << " bytes" << endl;
cout << "float: " << sizeof(discount) << " bytes" << endl;
cout << "double: " << sizeof(fareRupees) << " bytes" << endl;
cout << endl;
cout << "INT RANGE" << endl;
cout << "Minimum int: " << INT_MIN << endl;
cout << "Maximum int: " << INT_MAX << endl;
C:\Users\kavya\OneDrive\Doc
DATA TYPE SIZES
bool: 1 bytes
char: 1 bytes
int: 4 bytes
long long: 8 bytes
unsigned int: 4 bytes
float: 4 bytes
double: 8 bytes
INT RANGE
Minimum int: -2147483648
Maximum int: 2147483647
NETWORK COLLECTION
Journeys: 3000000
Fare per journey: Rs 30
Total in paise: 9000000000
Total in rupees: 90000000
TYPE COMPATIBILITY
Integer division: 3
Double division: 3.5
Integer converted to bool: true
MONEY PRECISION TEST
Double total in rupees: 100000.000001333
Integer paise total in rupees: 100000
Process exited after 0.5187 seconds with return value 0
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task3.exe
- Output Size: 1.83578491210938 MiB
- Compilation Time: 0.45s


C:\Users\kavya\OneDrive\Documents\task4.cpp - [Executing] - Dev-C++ 5.11
#include <iostream>
using namespace std;
int main()
{
double c = a / b;
double d = static_cast<double>(a) / b;
cout << c << ' ' << d << '\n';
char ch = 'A' + 2;
cout << ch << ' ' << ch + 1 << '\n';
bool flag = 5;
cout << flag << ' ' << boolalpha << flag << '\n';
unsigned int u = 1;
int s = -1;
cout << (s < u ? "s < u" : "s >= u") << '\n';
return 0;
}
C:\Users\kavya\OneDrive\Doc
3 3.5
C 68
1 true
s >= u
Process exited after 0.3831 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task4.exe
- Output Size: 1.83343029022217 MiB
- Compilation Time: 0.48s


C:\Users\kavya\OneDrive\Documents\task5.cpp - [Executing] - Dev-C++ 5.11
#include <iostream>
using namespace std;
int main()
{
int x = 5, y = 3;
cout << x + y * 2 << ' ' << (x + y) * 2 << '\n';
int z = x++ + ++y;
cout << z << ' ' << x << ' ' << y << '\n';
cout << (x > y ? x : y) << ' ' << x % 4 * 2 << '\n';
cout << 10 / 3 << ' ' << 10 % 3 << ' '
<< -10 / 3 << ' ' << -10 % 3 << '\n';
int k = 0;
for (int i = 0; i < 10; i += 3)
k += i;
cout << k << '\n';
return 0;
}
C:\Users\kavya\OneDrive\Doc
11 16
9 6 4
6 4
3 1 -3 -1
18
Process exited after 0.3352 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task5.exe
- Output Size: 1.83238887786865 MiB
- Compilation Time: 0.50s

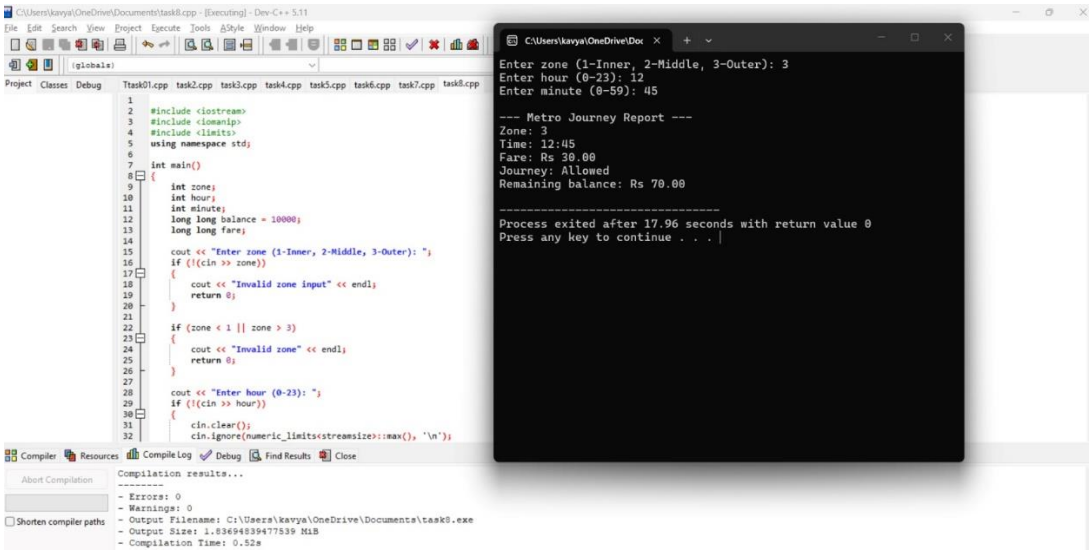

C:\Users\kavya\OneDrive\Documents\task6.cpp - [Executing] - Dev-C++ 5.11
#include <iostream>
#include <iomanip>
using namespace std;
enum Zone { Inner, Middle, Outer };
const int BASE_PAISE[] = {1000, 2000, 3000};
const int PEAK_SURCHARGE_PERCENT = 20;
const long long MIN_BALANCE_PAISE = 5000;
long long fare(Zone z, bool peak)
{
long long baseFare;
switch (z)
{
case Inner:
baseFare = BASE_PAISE[0];
break;
case Middle:
baseFare = BASE_PAISE[1];
break;
case Outer:
baseFare = BASE_PAISE[2];
break;
default:
return -1;
}
C:\Users\kavya\OneDrive\Doc
Fare Rs 10.00 balance Rs 90
Fare Rs 36.00 balance Rs 54
Denied (low balance) balance Rs 54
Process exited after 0.3581 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task
- Output Size: 1.83348178863525 MiB
- Compilation Time: 0.41s



```
PROCEDURAL VERSION
Journey allowed
Balance: Rs 90

OBJECT-ORIENTED VERSION
Journey allowed
Balance: Rs 64

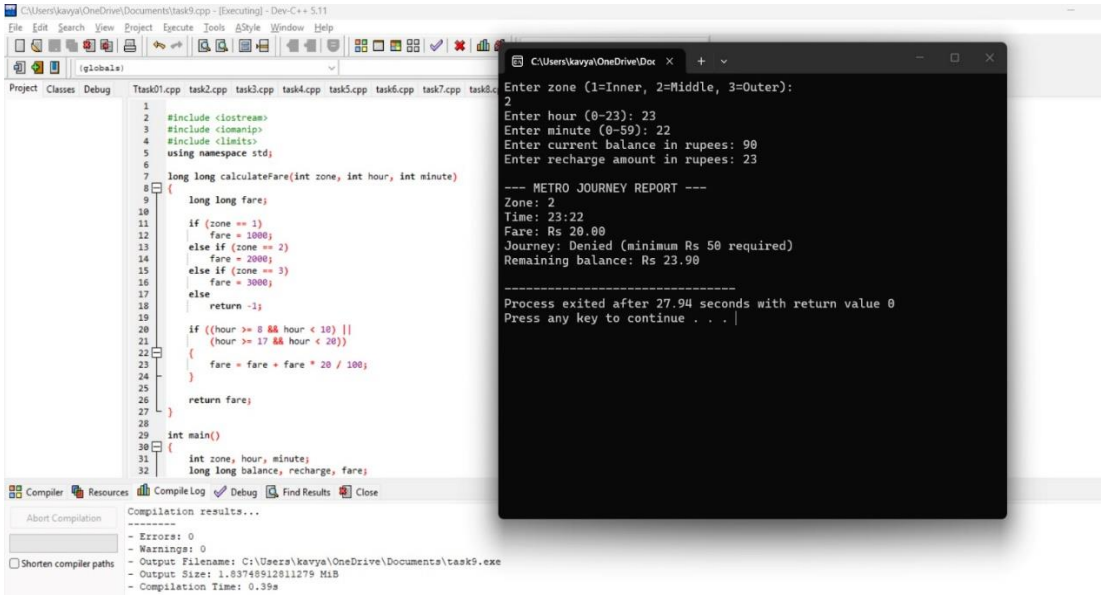
--------------------------------
Process exited after 0.3309 seconds with return value 0
Press any key to continue . . .
```



```
Enter zone (1-Inner, 2-Middle, 3-Outer): 3
Enter hour (0-23): 12
Enter minute (0-59): 45

--- Metro Journey Report ---
Zone: 3
Time: 12:45
Fare: Rs 30.00
Journey: Allowed
Remaining balance: Rs 70.00

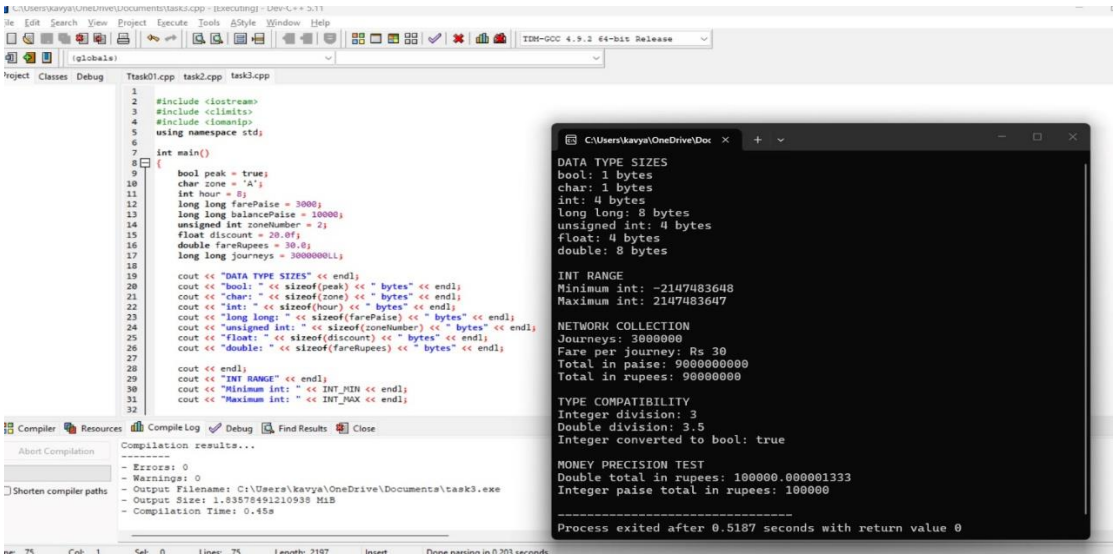
--------------------------------
Process exited after 17.96 seconds with return value 0
Press any key to continue . . .
```



```
Enter zone (1=Inner, 2=Middle, 3=Outer):
2
Enter hour (0-23): 23
Enter minute (0-59): 22
Enter current balance in rupees: 90
Enter recharge amount in rupees: 23

--- METRO JOURNEY REPORT ---
Zone: 2
Time: 23:22
Fare: Rs 20.00
Journey: Denied (minimum Rs 50 required)
Remaining balance: Rs 23.90

--------------------------------
Process exited after 27.94 seconds with return value 0
Press any key to continue . . .
```



```
#include <iostream>
#include <climits>
#include <iomanip>
using namespace std;

int main()
{
    bool peak = true;
    char zone = 'A';
    int hour = 8;
    long long farePaise = 3000;
    long long balancePaise = 10000;
    unsigned int zoneNumber = 2;
    float discount = 20.0f;
    double fareRupees = 30.0;
    long long journeys = 3000000LL;

    cout << "DATA TYPE SIZES" << endl;
    cout << "bool: " << sizeof(peak) << " bytes" << endl;
    cout << "char: " << sizeof(zone) << " bytes" << endl;
    cout << "int: " << sizeof(hour) << " bytes" << endl;
    cout << "long long: " << sizeof(farePaise) << " bytes" << endl;
    cout << "unsigned int: " << sizeof(zoneNumber) << " bytes" << endl;
    cout << "float: " << sizeof(discount) << " bytes" << endl;
    cout << "double: " << sizeof(fareRupees) << " bytes" << endl;

    cout << endl;
    cout << "INT RANGE" << endl;
    cout << "Minimum int: " << INT_MIN << endl;
    cout << "Maximum int: " << INT_MAX << endl;
```

```
DATA TYPE SIZES
bool: 1 bytes
char: 1 bytes
int: 4 bytes
long long: 8 bytes
unsigned int: 4 bytes
float: 4 bytes
double: 8 bytes

INT RANGE
Minimum int: -2147483648
Maximum int: 2147483647

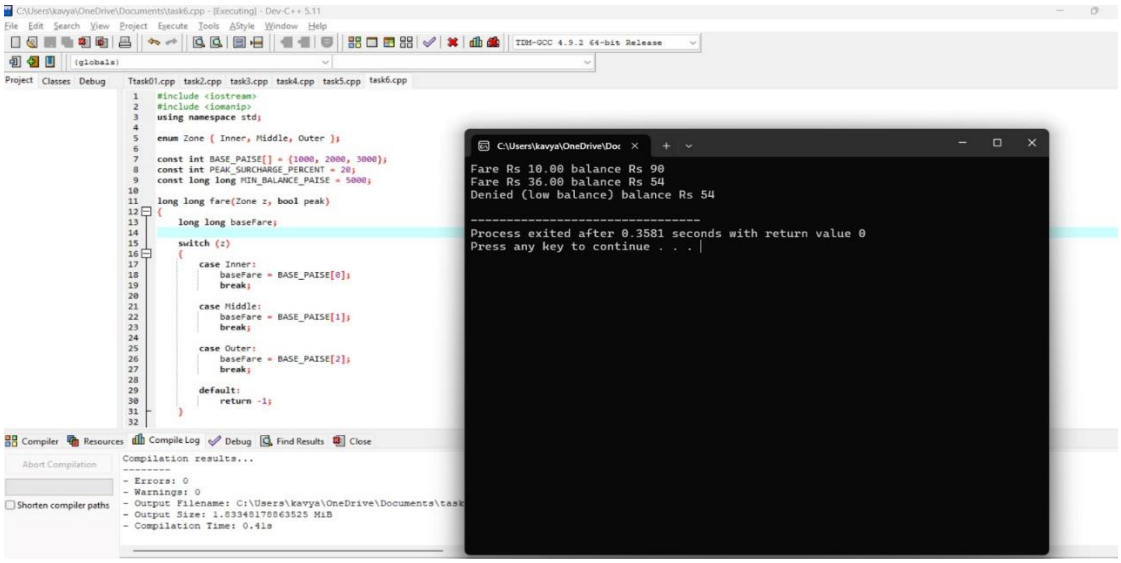
NETWORK COLLECTION
Journeys: 3000000
Fare per journey: Rs 30
Total in paise: 9000000000
Total in rupees: 90000000

TYPE COMPATIBILITY
Integer division: 3
Double division: 3.5
Integer converted to bool: true

MONEY PRECISION TEST
Double total in rupees: 100000.000001333
Integer paise total in rupees: 100000

--------------------------------
Process exited after 0.5187 seconds with return value 0
```

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task3.exe
- Output Size: 1.835784912109375 MiB
- Compilation Time: 0.45s
```



```
#include <iostream>
#include <iomanip>
using namespace std;

enum Zone { Inner, Middle, Outer };

const int BASE_PAISE[] = {1000, 2000, 3000};
const int PEAK_SURCHARGE_PERCENT = 20;
const long long MIN_BALANCE_PAISE = 5000;

long long fare(Zone z, bool peak)
{
    long long baseFare;

    switch (z)
    {
        case Inner:
            baseFare = BASE_PAISE[0];
            break;

        case Middle:
            baseFare = BASE_PAISE[1];
            break;

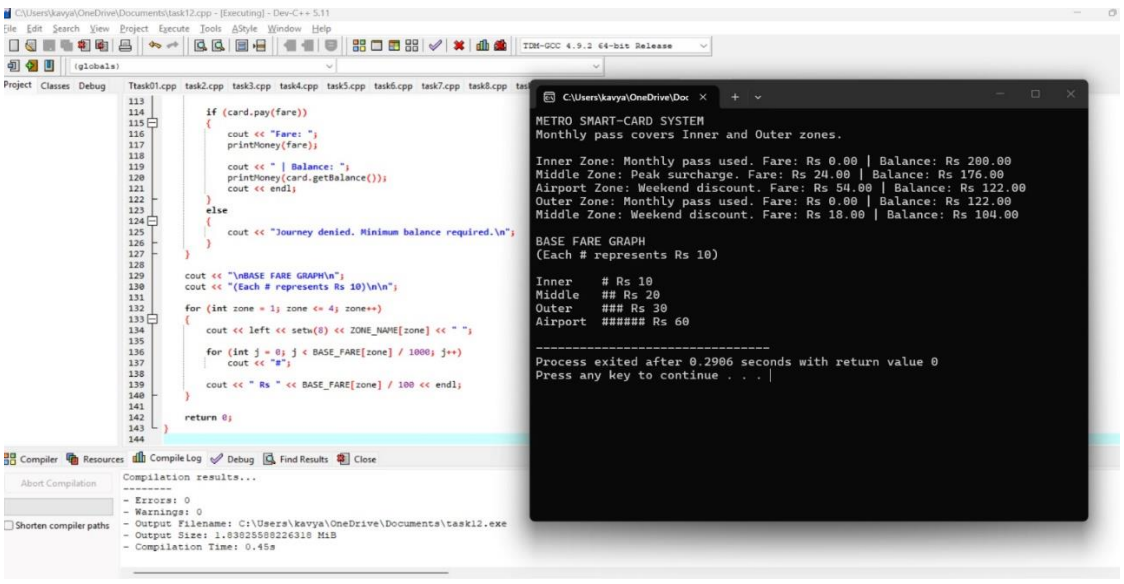
        case Outer:
            baseFare = BASE_PAISE[2];
            break;

        default:
            return -1;
    }
```

```
Fare Rs 10.00 balance Rs 90
Fare Rs 36.00 balance Rs 54
Denied (low balance) balance Rs 54

--------------------------------
Process exited after 0.3581 seconds with return value 0
Press any key to continue . . .
```

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task
- Output Size: 1.83348178863525 MiB
- Compilation Time: 0.41s
```



```
        if (card.pay(fare))
        {
            cout << "Fare: ";
            printMoney(fare);

            cout << " | Balance: ";
            printMoney(card.getBalance());
            cout << endl;
        }
        else
        {
            cout << "Journey denied. Minimum balance required.\n";
        }
    }

    cout << "\nBASE FARE GRAPH\n";
    cout << "(Each # represents Rs 10)\n\n";

    for (int zone = 1; zone <= 4; zone++)
    {
        cout << left << setw(8) << ZONE_NAME[zone] << " ";

        for (int j = 0; j < BASE_FARE[zone] / 1000; j++)
            cout << "#";

        cout << " Rs " << BASE_FARE[zone] / 100 << endl;
    }

    return 0;
}
```

```
METRO SMART-CARD SYSTEM
Monthly pass covers Inner and Outer zones.

Inner Zone: Monthly pass used. Fare: Rs 0.00 | Balance: Rs 200.00
Middle Zone: Peak surcharge. Fare: Rs 24.00 | Balance: Rs 176.00
Airport Zone: Weekend discount. Fare: Rs 54.00 | Balance: Rs 122.00
Outer Zone: Monthly pass used. Fare: Rs 0.00 | Balance: Rs 122.00
Middle Zone: Weekend discount. Fare: Rs 18.00 | Balance: Rs 104.00

BASE FARE GRAPH
(Each # represents Rs 10)

Inner    # Rs 10
Middle   ## Rs 20
Outer    ### Rs 30
Airport  ###### Rs 60

--------------------------------
Process exited after 0.2906 seconds with return value 0
Press any key to continue . . .
```

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kavya\OneDrive\Documents\task12.exe
- Output Size: 1.838255882263184 MiB
- Compilation Time: 0.45s
```